

Christopher P. Ravosa

Boston, MA (Open to Remote) | christopher.ravosa99@gmail.com

Website: <https://www.chrisravosa.com> | **LinkedIn:** <https://www.linkedin.com/in/christopherravosa/>

SUMMARY

Software engineer with experience building distributed backend systems, AI infrastructure, simulation platforms, and real-time applications across cloud and gaming technologies.

SKILLS

Languages:	C#, TypeScript, Python, Java, JavaScript, Rust, C++, SQL
Databases & Mappers:	PostgreSQL, MongoDB, SQLAlchemy, Mongoose
Frameworks & Runtimes:	Node.js, Express, Next.js, React, React Native, FastAPI
Systems & Architecture:	REST APIs, Distributed Systems, State Machines, WebSockets, TCP Networking, CI/CD
Cloud & DevOps:	Git, Docker, Perforce, GCP, AWS
Game Engines:	Unity, Godot, Unreal

PROFESSIONAL EXPERIENCE

Software Engineer - Moku

Mar 2025 - June 2025

- **Reduced AI model iteration cycle by 60 days** by designing and building a new reinforcement learning infrastructure
- Led design and implementation of systems to manage parallel Unity simulations for training and evaluating AI agents
- Provisioned Linux VM infrastructure on Google Compute Engine for AI simulation and debugging
- Built a deterministic replay pipeline for debugging, auditing, and reproducing execution
- Utilized Python, C#, JSON and YAML configuration files to create runtime configurable agent training environments
- Coordinated engineering efforts to implement gameplay systems for 2D and 3D Unity games

Associate Software Engineer - Major League Baseball

Aug 2023 - Mar 2025

- Led development of an interactive software product when lead engineer was reallocated to another project
- Designed, authored, and deployed backend APIs supporting two live Unity applications
- Developed and owned real-time TCP multiplayer servers supporting live user interactions
- Built pipelines for distributing assets via GCP CDN, enabling dynamic content delivery while reducing client footprint
- **Diagnosed and resolved production outage in 15 minutes** while participating in on-call rotations
- Created server stress testing frameworks for identifying integrity issues and improving reliability under sustained load
- **Reduced application memory usage by 40%** by developing compute shaders in HLSL to generate textures at runtime
- Authored spatial computing systems to visualize real-time baseball events in 3D space for MLB on Apple Vision Pro

Associate Technical Designer - Naughty Dog

Jul 2022 - Jun 2023

- Built networked systems using a host-authoritative architecture, ensuring state consistency across distributed clients
- Implemented synchronization logic resilient to latency, packet loss, and unreliable network conditions
- Designed and tested systems to prevent state divergence and maintain consistency in real-time environments

PROJECTS

Real-Time Climate Monitoring System

- Built an IoT device and fault-tolerant, climate monitoring firmware for an ESP32 microcontroller using ESP-IDF and C
- Implemented bidirectional WebSocket communication between firmware, cloud-hosted backend, and browser clients
- Designed and built a web dashboard for end users to view data and remotely control the IoT device

Chasm Master

- Built an **LLM-powered semantic validation service** that maps free-form player input to predefined game actions
- Automated server deployment using GitHub Actions to support a **playable demo on ChrisRavosa.com**

EDUCATION

Marist College

- **MSc in Software Development**, Concentration in **Mobile Computing** | Sept. 2020 to May 2022
- **BSc in Computer Science** | Sept. 2017 to May 2021 (**GPA 3.87**)